Problem · Unit 5 · 10 points · 60 min · Standard exam level

P10 — Free fall from a tower on the rotating Earth: eastward deflection, the order of the north-south effect and an upward shot

*A particle P of mass m is released **from rest relative to the ground** from the top of a tower of height h standing at the point A of the Earth's surface at latitude λ (northern hemisphere). The Earth rotates with constant angular velocity of modulus Ω about its polar axis, whose unit vector $\mathbf{k}_E$ points to the north pole; the centre of the Earth is taken as fixed in an inertial frame S_0.*

*The local frame $S_1 = \{A; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is attached to the ground at the foot A of the tower and rotates with the Earth: x points **east**, y points **north**, z points **up** along the plumb line. In S_1 the particle feels the effective gravity $-g\,\mathbf{k}$, which already contains the centrifugal force of the point A (gravitation plus $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}_{CA})$, C the centre of the Earth). Air drag is neglected and h is small compared with the radius of the Earth.*

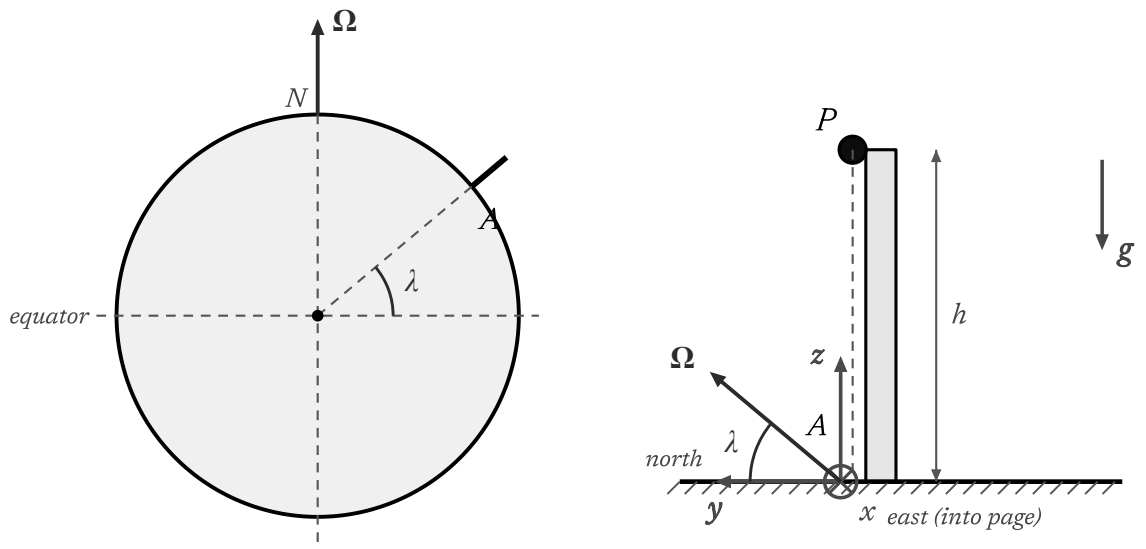


Figure 1. Free fall on the rotating Earth: point A at latitude λ, rotation $\boldsymbol{\Omega}$ along the Earth axis; local frame $S_1 = \{A; x\ \text{east}, y\ \text{north}, z\ \text{up}\}$ (east points into the page), $\boldsymbol{\Omega} = \Omega\cos\lambda\,\mathbf{j} + \Omega\sin\lambda\,\mathbf{k}$, particle P released from rest at height h.

Setup

- Data: m, h, g, Ω, λ. Numerical part: $h = 100$ m, $\lambda = 40^\circ$, $g = 9.81$ m s^{-2}, $\Omega = 7.292 \times 10^{-5}$ rad s^{-1} (2π per sidereal day of 86 164 s).
- Hypotheses: point particle, uniform effective gravity along the height of the tower, Ω constant, $\Omega t \ll 1$ during the fall (so a solution to first order in Ω is meaningful), no drag.
- S_1 is a **rotating** frame: $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_E$ with $\dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$.

Part (a) · 1.5 points

Degrees of freedom, angular velocity of S_1 and inertia forces

How many degrees of freedom has P? Write $\boldsymbol{\omega}_{10}$ in the basis of S_1 and list the inertia forces that act on P in S_1, saying which one is absorbed into $-g\,\mathbf{k}$ and which one is kept.

Step 1. Count

P is a free particle in space: 3 degrees of freedom, the coordinates (x, y, z) of $\mathbf{r} = \overrightarrow{AP}$ in S_1. Nothing constrains it between release and impact.

Step 2. The angular velocity in the local basis

The local vertical $\mathbf{k}$ is the plumb line at A (statement), and the latitude λ is measured from this plumb line: it is the angle between $\mathbf{k}$ and the equatorial plane. The angle between $\mathbf{k}$ and $\mathbf{k}_E$ is therefore $90^\circ - \lambda$, so $\mathbf{k}_E \cdot \mathbf{k} = \sin\lambda$ by definition of λ. The polar axis lies in the meridian plane through A, which contains $\mathbf{j}$ (north) and $\mathbf{k}$, so $\mathbf{k}_E \cdot \mathbf{i} = 0$, and it leans towards the north: $\mathbf{k}_E \cdot \mathbf{j} = \cos\lambda$. Check: $\cos^2\lambda + \sin^2\lambda = 1$. The ground turns with the Earth, so

$$\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_E = \Omega\cos\lambda\,\mathbf{j} + \Omega\sin\lambda\,\mathbf{k}, \qquad \dot{\boldsymbol{\omega}}_{10} = \mathbf{0}$$

Step 3. Change of basis

The polar axis is best described with the cylindrical basis of the Earth at A, $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{k}_E\}$: $\mathbf{e}_\rho$ perpendicular to the polar axis and pointing away from it, in the meridian plane of A, $\mathbf{e}_\varphi = \mathbf{i}$ (east), $\mathbf{k}_E$ the polar axis. This basis turns about $\mathbf{k}_E$ with angular velocity Ω relative to S_0, like the local one. Projecting the local unit vectors ($\mathbf{j}\cdot\mathbf{e}_\rho = -\sin\lambda$, $\mathbf{j}\cdot\mathbf{k}_E = \cos\lambda$, $\mathbf{k}\cdot\mathbf{e}_\rho = \cos\lambda$, $\mathbf{k}\cdot\mathbf{k}_E = \sin\lambda$):

$$\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\\ \mathbf{k}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{e}_\rho\\ \mathbf{e}_\varphi\\ \mathbf{k}_E\end{pmatrix}, \qquad [R] = \begin{pmatrix}0 & 1 & 0\\ -\sin\lambda & 0 & \cos\lambda\\ \cos\lambda & 0 & \sin\lambda\end{pmatrix}$$

Checks: the rows have unit length (1, $\sin^2\lambda+\cos^2\lambda$, $\cos^2\lambda+\sin^2\lambda$) and are mutually orthogonal (row 1 with rows 2 and 3: 0; rows 2 and 3: $-\sin\lambda\cos\lambda+\cos\lambda\sin\lambda=0$), so $[R][R]^T=I$; expanding along the first row, $\det[R]=-1\cdot\big((-\sin\lambda)\sin\lambda-\cos\lambda\cos\lambda\big)=+1$. The third column of $[R]$ gives back $\mathbf{k}_E=\cos\lambda\,\mathbf{j}+\sin\lambda\,\mathbf{k}$.

Step 4. Inertia forces (Notes §3.2 and §4.9)

The acceleration of P in S_0 is $\mathbf{a}=\mathbf{a}_{rel}+\mathbf{a}_{drag}+2\boldsymbol{\omega}_{10}\times\mathbf{v}_{rel}$. The point of S_1 that coincides with P moves on a circle about the polar axis (the centre C is fixed and $\boldsymbol{\omega}_{10}$ constant), so $\mathbf{a}_{drag}=\boldsymbol{\Omega}\times\big(\boldsymbol{\Omega}\times\mathbf{r}_{CP}\big)$ with $\mathbf{r}_{CP}=\mathbf{r}_{CA}+\mathbf{r}$. In S_1, $m\mathbf{a}_{rel}=\mathbf{F}+\mathbf{F}_{cf}+\mathbf{F}_{cor}$ with

$$\mathbf{F}_{cf}=-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}_{CA})-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}),\qquad\mathbf{F}_{cor}=-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel}$$

The first term is the same at every point of the tower: added to the gravitational attraction it gives the effective weight $-mg\,\mathbf{k}$ that a plumb line at A measures (this is how $\mathbf{k}$ is defined). The second term has modulus at most $m\Omega^2h\approx5.3\times10^{-7}\,m$ m s^{-2} for $h=100$ m: it is of order Ω^2 and is dropped. The Coriolis force is of order Ω and is the one that matters.

KEY RESULTS

- 3 degrees of freedom (x,y,z), no constraint during the fall.
- $\boldsymbol{\omega}_{10}=\Omega\cos\lambda\,\mathbf{j}+\Omega\sin\lambda\,\mathbf{k}$, constant.
- Inertia forces: centrifugal $-m\boldsymbol{\Omega}\times\big(\boldsymbol{\Omega}\times(\mathbf{r}_{CA}+\mathbf{r})\big)$ (the part with $\mathbf{r}_{CA}$ is in g; the part with $\mathbf{r}$ is $O(\Omega^2h)$ and is neglected) and Coriolis $-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel}$ (kept). No Euler force.

Part (b) · 2 points

Equations of motion in S_1

Write the vector equation of motion of P in S_1 and its three scalar components, with the initial conditions.

Step 1. Vector equation

With the results of (a), the only forces in S_1 are the effective weight and the Coriolis force:

$$m\,\ddot{\mathbf{r}}=-mg\,\mathbf{k}-2m\,\boldsymbol{\Omega}\times\dot{\mathbf{r}}$$

Step 2. The cross product

$$\boldsymbol{\Omega}\times\dot{\mathbf{r}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & \Omega\cos\lambda & \Omega\sin\lambda \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix} = \Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big)\mathbf{i} + \Omega\dot{x}\sin\lambda\,\mathbf{j} - \Omega\dot{x}\cos\lambda\,\mathbf{k}$$

Step 3. Components

Dividing by m and projecting on $\mathbf{i},\mathbf{j},\mathbf{k}$:

$$\ddot{x} = -2\Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big), \qquad \ddot{y} = -2\Omega\dot{x}\sin\lambda, \qquad \ddot{z} = -g + 2\Omega\dot{x}\cos\lambda$$

The particle starts at the top of the tower with zero velocity **relative to the ground** (in S_0 it moves east with the tower top): $\mathbf{r}(0) = h\,\mathbf{k}$, $\dot{\mathbf{r}}(0) = \mathbf{0}$.

KEY RESULTS

- $\ddot{x} = -2\Omega\big(\dot{z}\cos\lambda - \dot{y}\sin\lambda\big),\ \ddot{y} = -2\Omega\dot{x}\sin\lambda,\ \ddot{z} = -g + 2\Omega\dot{x}\cos\lambda.$
- $x(0) = y(0) = 0, z(0) = h, \dot{x}(0) = \dot{y}(0) = \dot{z}(0) = 0.$

Part (c) · 2 points

Solution to first order in Ω

Solve the equations to first order in Ω and show that the particle lands at $\Delta x = \dfrac{2\sqrt{2}}{3}\,\Omega\cos\lambda\sqrt{\dfrac{h^3}{g}}$ east of the foot of the tower. Does the time of fall change at this order?

Step 1. Zeroth order

With $\Omega = 0$: $\ddot{x}_0 = \ddot{y}_0 = 0$ and $\ddot{z}_0 = -g$, with $\dot{\mathbf{r}}_0(0) = \mathbf{0}$ and $\mathbf{r}_0(0) = h\,\mathbf{k}$. Integrating twice with the limits 0 and t (antiderivative, then Barrow's rule $[F]_0^t = F(t) - F(0)$):

$$\dot{z}_0(t) = \dot{z}_0(0) + \int_0^t (-g)\,d\tau = 0 + \big[-g\tau\big]_0^t = -gt - (-g\cdot 0) = -gt$$

$$z_0(t) = z_0(0) + \int_0^t (-g\tau)\,d\tau = h + \Big[-\tfrac{1}{2}g\tau^2\Big]_0^t = h - \tfrac{1}{2}gt^2$$

$$\dot{x}_0(t) = \dot{x}_0(0) + \int_0^t 0\,d\tau = 0, \qquad x_0(t) = x_0(0) + \int_0^t 0\,d\tau = 0$$

and in the same way $\dot{y}_0 = 0$, $y_0 = 0$.

Step 2. First order

Every Coriolis term carries a factor Ω, so to first order the velocity inside it is the zeroth-order one, $\dot{\mathbf{r}}_0 = -gt\,\mathbf{k}$:

$$\ddot{x}_1 = -2\Omega\big(\dot{z}_0\cos\lambda - \dot{y}_0\sin\lambda\big) = 2\Omega g t\cos\lambda, \qquad \ddot{y}_1 = -2\Omega\dot{x}_0\sin\lambda = 0, \qquad \ddot{z}_1 = 2\Omega\dot{x}_0\cos\lambda = 0$$

with zero initial conditions, so $y_1 = z_1 = 0$ and

$$\dot{x}_1 = \int_0^t 2\Omega g\cos\lambda\,\tau\,d\tau = \big[\Omega g\cos\lambda\,\tau^2\big]_0^t = \Omega g t^2\cos\lambda$$

$$x_1 = \int_0^t \Omega g\cos\lambda\,\tau^2\,d\tau = \Big[\tfrac{1}{3}\Omega g\cos\lambda\,\tau^3\Big]_0^t = \tfrac{1}{3}\Omega g t^3\cos\lambda$$

Step 3. Landing point

Since $z_1 = 0$, the height is $h - \tfrac{1}{2}gt^2$ up to $O(\Omega^2)$ and the particle reaches $z = 0$ at $t_f = \sqrt{2h/g}$, as without rotation. Then $t_f^3 = (2h/g)^{3/2} = 2\sqrt{2}\,h^{3/2}/g^{3/2}$ and

$$\Delta x = x_1(t_f) = \tfrac{1}{3}\Omega g\cos\lambda\cdot\frac{2\sqrt{2}\,h^{3/2}}{g^{3/2}} = \frac{2\sqrt{2}}{3}\,\Omega\cos\lambda\sqrt{\frac{h^3}{g}} > 0$$

The particle lands **east** of the foot A.

Step 4. Physical reading in S_0

At release the top of the tower is a distance $h\cos\lambda$ farther from the polar axis than the foot, so in S_0 it moves east $\Omega h\cos\lambda$ faster **than the foot**. The particle keeps that eastward speed while it falls, which is why it ends up ahead of the foot. But this lead speed $\Omega h\cos\lambda$ is measured against the foot, not against the rotating ground under the particle: relative to the ground (S_1) the particle starts at rest, and its eastward speed $\dot{x}_1 = \Omega g t^2\cos\lambda$ is zero at release and grows like t^2, as the particle descends to points of the ground that move more slowly. The crude estimate "lead speed $\Omega h\cos\lambda$ during the whole fall", $\Omega h\cos\lambda\, t_f$, assumes the full lead from the start and is therefore $3/2$ of the exact first-order value: with $h = \tfrac{1}{2}gt_f^2$, $\Omega h\cos\lambda\, t_f = \tfrac{1}{2}\Omega g\cos\lambda\, t_f^3$, against $\Delta x = \tfrac{1}{3}\Omega g\cos\lambda\, t_f^3$.

KEY RESULTS

- Order 0: $z_0 = h - \tfrac{1}{2}gt^2$, $\dot{z}_0 = -gt$, $x_0 = y_0 = 0$.
- Order 1: $\ddot{x}_1 = 2\Omega g t\cos\lambda,\ \dot{x}_1 = \Omega g t^2\cos\lambda,\ x_1 = \tfrac{1}{3}\Omega g t^3\cos\lambda$; $y_1 = z_1 = 0$.
- $t_f = \sqrt{2h/g}$ (unchanged at first order), $\Delta x = \tfrac{2\sqrt{2}}{3}\Omega\cos\lambda\sqrt{h^3/g}$, to the **east**.

Part (d) · 1 point

Numbers

For $h = 100$ m, $\lambda = 40^\circ$: compute the time of fall and the eastward deflection. Where is the particle (east of the plumb line) at $t_f/2$?

Step 1. Time and deflection

$t_f = \sqrt{2\cdot 100/9.81} = \sqrt{20.39} = 4.515$ s. With $\cos 40^\circ = 0.7660$ and $\sqrt{h^3/g} = \sqrt{10^6/9.81} = 319.3$ s:

$$\Delta x = \frac{2\sqrt{2}}{3}\,(7.292\times 10^{-5})(0.7660)(319.3)\ \text{m} = 1.681\times 10^{-2}\ \text{m} = 1.681\ \text{cm}$$

A numerical integration of the exact equations of (b) (no expansion) gives the same value to four significant figures.

Step 2. Half-way

$x_1 \propto t^3$, so at $t_f/2$ the deflection is $(1/2)^3\Delta x = 0.210$ cm: most of the deflection builds up in the second half of the fall.

KEY RESULTS

- $t_f = \sqrt{200/9.81} = 4.515$ s.
- $\Delta x = 1.681$ cm east; $x_1(t_f/2) = \Delta x/8 = 0.210$ cm.

Part (e) · 1.5 points

Is there a north-south deflection?

A student integrates the y equation once more and reports a southward deflection $y_2(t_f) = -\tfrac{2}{3}\Omega^2 h^2 \sin\lambda\cos\lambda/g$. Check the formula, and say whether it is a consistent result of the model of the statement.

Step 1. The second-order term

At second order the y equation receives $\dot{x}_1$: $\ddot{y}_2 = -2\Omega\dot{x}_1\sin\lambda = -2\Omega^2 g t^2 \sin\lambda\cos\lambda$. Integrating twice from rest:

$$\dot{y}_2(t) = \int_0^t \left(-2\Omega^2 g\sin\lambda\cos\lambda\,\tau^2\right)d\tau = \left[-\tfrac{2}{3}\Omega^2 g\sin\lambda\cos\lambda\,\tau^3\right]_0^t = -\tfrac{2}{3}\Omega^2 g t^3\sin\lambda\cos\lambda$$

$$y_2(t) = \int_0^t \left(-\tfrac{2}{3}\Omega^2 g\sin\lambda\cos\lambda\,\tau^3\right)d\tau = \left[-\tfrac{1}{6}\Omega^2 g\sin\lambda\cos\lambda\,\tau^4\right]_0^t = -\tfrac{1}{6}\Omega^2 g t^4\sin\lambda\cos\lambda$$

(the lower limit contributes 0 in both, since $\dot{y}_2(0) = y_2(0) = 0$).

and with $t_f^4 = 4h^2/g^2$: $y_2(t_f) = -\tfrac{2}{3}\Omega^2 h^2\sin\lambda\cos\lambda/g$. With the data: -1.78×10^{-6} m, about 10^{-4} of Δx (the ratio is $\tfrac{1}{2}\Omega t_f\sin\lambda$).

Step 2. Why it is not a consistent answer

The equation of (b) is only exact to first order in Ω: in (a) we dropped the residual centrifugal force $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$, which is $O(\Omega^2)$. For $\mathbf{r}\approx z\,\mathbf{k}$ its y component is $-m\Omega^2 z\sin\lambda\cos\lambda$; with $z = z_0(t)$ it alone displaces the particle by

$$y_{cf}(t_f) = -\Omega^2\sin\lambda\cos\lambda\int_0^{t_f}\!\!\int_0^{\tau} z_0(s)\,ds\,d\tau = -\tfrac{5}{6}\,\frac{\Omega^2 h^2\sin\lambda\cos\lambda}{g} = -2.22\times10^{-6}\ \text{m}$$

which is larger than y_2. The direction of the gravitational attraction also changes along the vertical through A by an amount of the same order. A second-order answer must collect all these terms; keeping only the Coriolis one is inconsistent. **To first order in Ω, the only deflection is the eastward one**, and $y = O(\Omega^2)$ is at the micrometre level.

KEY RESULTS

- $\ddot{y}_2 = -2\Omega \dot{x}_1 \sin\lambda = -2\Omega^2 g t^2 \sin\lambda\cos\lambda$, $y_2 = -\tfrac{1}{6}\Omega^2 g t^4 \sin\lambda\cos\lambda$; at t_f: $-\tfrac{2}{3}\Omega^2 h^2 \sin\lambda\cos\lambda/g = -1.78\times10^{-6}$ m.
- The formula is right but **not consistent**: it is $O(\Omega^2)$, the same order as the residual centrifugal term $-m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r})$ dropped in (a), whose own effect is $-\tfrac{5}{6}\Omega^2 h^2 \sin\lambda\cos\lambda/g = -2.22\times10^{-6}$ m, and as the change of direction of gravitation along the fall. To first order in Ω there is no north-south deflection.

Part (f) · 2 points

A projectile fired vertically upward

From A a projectile is fired vertically upward (along $\mathbf{k}$) with speed v_0 relative to the ground. To first order in Ω, where does it land? Compute it for $v_0 = 50\ m\ s^{-1}$ at $\lambda = 40^\circ$, and compare with the drop of (c) from the same maximum height.

Step 1. Zeroth and first order

Order 0: $z_0 = v_0 t - \tfrac{1}{2}gt^2$, $\dot{z}_0 = v_0 - gt$, back at $z=0$ at $T = 2v_0/g$. Order 1, from the x equation of (b) with $\dot{y}_0 = 0$:

$$\ddot{x}_1 = -2\Omega\cos\lambda\,(v_0 - gt)$$

$$\dot{x}_1 = \int_0^t -2\Omega\cos\lambda\,(v_0 - g\tau)\,d\tau = -2\Omega\cos\lambda\Big[v_0\tau - \tfrac{1}{2}g\tau^2\Big]_0^t = -\Omega g\cos\lambda\, t\Big(\frac{2v_0}{g} - t\Big)$$

$$x_1 = \int_0^t -2\Omega\cos\lambda\Big(v_0\tau - \tfrac{1}{2}g\tau^2\Big)d\tau = -2\Omega\cos\lambda\Big[\tfrac{1}{2}v_0\tau^2 - \tfrac{1}{6}g\tau^3\Big]_0^t = -\Omega\cos\lambda\Big(v_0 t^2 - \tfrac{1}{3}gt^3\Big)$$

Step 2. Landing point

At $T = 2v_0/g$: $v_0 T^2 = 4v_0^3/g^2$ and $\tfrac{1}{3}gT^3 = \tfrac{8}{3}v_0^3/g^2$, so

$$\Delta x = x_1(T) = -\Omega\cos\lambda\Big(4 - \tfrac{8}{3}\Big)\frac{v_0^3}{g^2} = -\frac{4}{3}\,\Omega\cos\lambda\,\frac{v_0^3}{g^2} < 0$$

The projectile lands **west** of the launch point. With $v_0 = 50$ m s^{-1}: $\Delta x = -\frac{4}{3}(7.292 \times 10^{-5})(0.7660)(125\,000)/96.24$ m $= -9.674$ cm; the flight lasts $T = 10.19$ s and the apex is at $v_0^2/2g = 127.4$ m.

Step 3. Why it does not cancel

On the way up ($\dot z_0 > 0$) the Coriolis force pushes west; on the way down it pushes east. The symmetric-looking guess "the two halves cancel" confuses force with displacement: $\dot x_1 = -\Omega g \cos\lambda\, t(T-t)$ is negative during the whole flight and only returns to zero at $t = T$. The westward velocity gained on the way up is removed on the way down, but the westward **displacement** accumulated meanwhile remains.

Step 4. Comparison with the drop

Setting $v_0 = \sqrt{2gh}$ (apex at height h): $\Delta x_{up} = -\frac{4}{3}\Omega\cos\lambda(2gh)^{3/2}/g^2 = -4\cdot\frac{2\sqrt{2}}{3}\Omega\cos\lambda\sqrt{h^3/g}$, that is, -4 times the eastward deflection of (c): -6.73 cm for $h = 100$ m. The descent of the projectile alone would push it 1.68 cm east, but it starts the descent already moving west.

KEY RESULTS

- $\ddot x_1 = -2\Omega\cos\lambda\,(v_0 - gt),\ \dot x_1 = -\Omega g\cos\lambda\, t\,(T-t) \le 0,\ x_1 = -\Omega\cos\lambda\big(v_0 t^2 - \tfrac{1}{3}gt^3\big)$, with $T = 2v_0/g$.
- $\Delta x = -\frac{4}{3}\,\Omega\cos\lambda\, v_0^3/g^2$: it lands **west** of A; -9.674 cm for $v_0 = 50$ m s^{-1} ($T = 10.19$ s, apex 127.4 m).
- With $v_0 = \sqrt{2gh}$ the westward offset is 4 times the eastward deflection of the drop from h.

Interpretation and checks

- **Dimensional check:** $[\Omega g t^3] = \mathrm{s^{-1}}\cdot\mathrm{m\,s^{-2}}\cdot\mathrm{s^3} = \mathrm{m}$; $[\Omega\sqrt{h^3/g}] = \mathrm{s^{-1}}\sqrt{\mathrm{m^3/(m\,s^{-2})}} = \mathrm{s^{-1}}\cdot\mathrm{m\,s} = \mathrm{m}$; $[\Omega v_0^3/g^2] = \mathrm{s^{-1}\,m^3 s^{-3}/(m^2 s^{-4})} = \mathrm{m}$.
- **Limits:** at the poles ($\lambda = 90^\circ$) $\cos\lambda = 0$: $\boldsymbol{\Omega}$ is vertical, parallel to the fall, and there is no deflection; it is largest at the equator. In the southern hemisphere ($\lambda < 0$) $\cos\lambda > 0$ still: the drop also lands **east**. The expansion requires $\Omega t_f \ll 1$ (here 3.3×10^{-4}).
- **Common mistakes in the exam:** (1) writing $\boldsymbol{\Omega} = \Omega\,\mathbf{k}$ (the local vertical) instead of $\Omega(\cos\lambda\,\mathbf{j} + \sin\lambda\,\mathbf{k})$; (2) a sign error in the Coriolis force (it is $-2m\boldsymbol{\Omega}\times\mathbf{v}_{rel}$) that sends the particle west; (3) keeping $O(\Omega^2)$ Coriolis terms while dropping the centrifugal term of the same order; (4) assuming that the up and down halves of a vertical shot cancel.